General Physics – 1121TEIXB1S0290 2P

Introduction to Special Relativity

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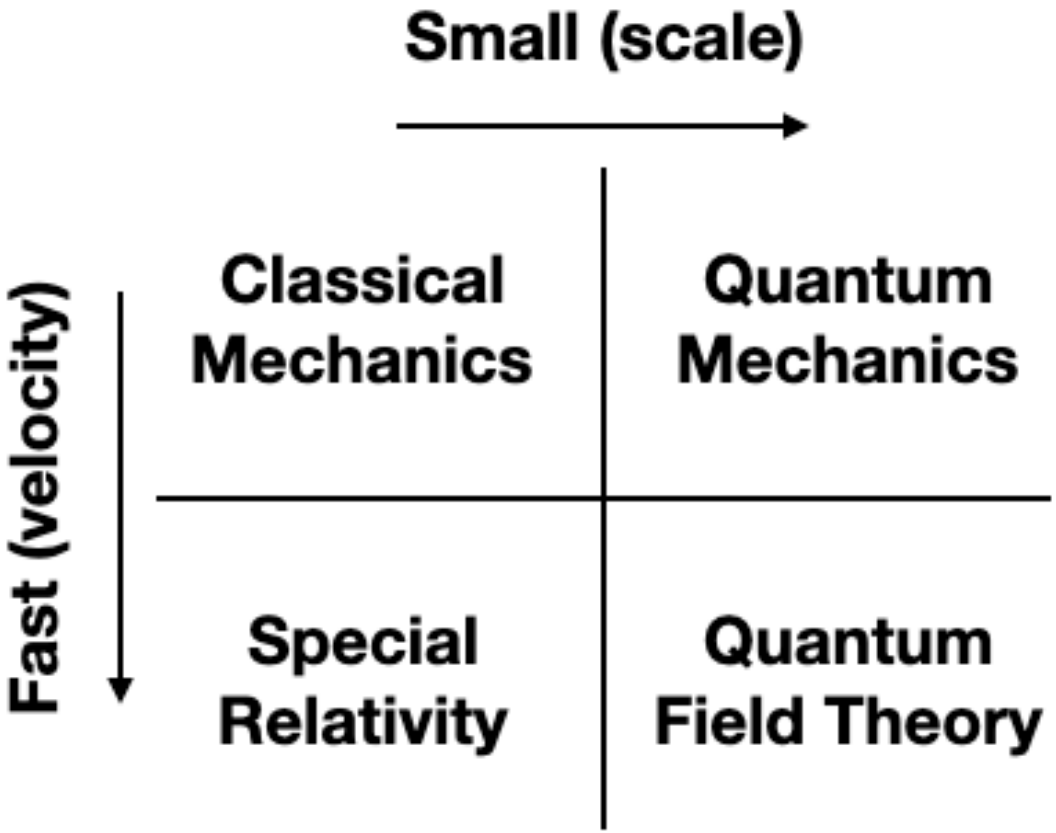
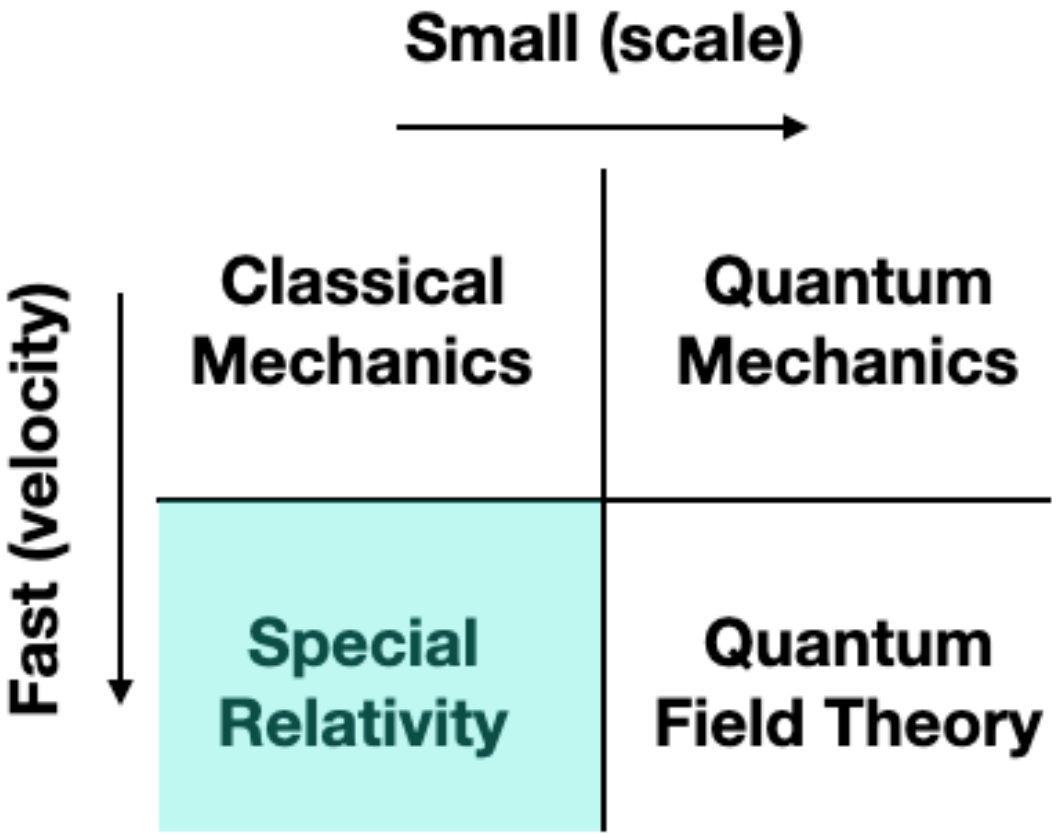
June 21, 2024@TKU

Physics with scale and velocity



Physics with scale and velocity

- Special relativity:
  - It is a *non-quantum* theory.
  - It describes high speed motion.



Outline

1 Physical Observation

2 Special Relativity

3 Physical Consequences

4 Relativistic Kinematics

5 Geometric Meaning

6 Lorentz Transformation

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Light and EM Waves

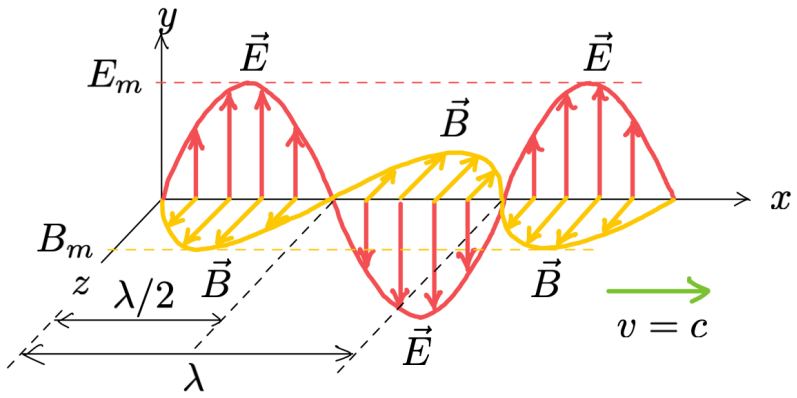
Maxwell Equatoins

- Gauss's law $\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$ (charges produce electric field $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$)
- Gauss's law for magnetism $\oint \vec{B} \cdot d\vec{A} = 0$ (no magnetic monopole $\vec{\nabla} \cdot \vec{B} = 0$)
- Faraday's law $\mathcal{E} = \oint \vec{E} \cdot d\vec{l} = -\frac{\partial \Phi_B}{\partial t}$ (changing magnetic flux produces electric field $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$)
- Ampére-Maxwell law $\oint \vec{B} \cdot d\vec{l} = \mu_0 (I + \underbrace{\epsilon_0 \frac{\partial \Phi_E}{\partial t}}_{\text{the } displacement\ current\ I_D})$ (current and changing electric flux produce magnetic field $\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$)

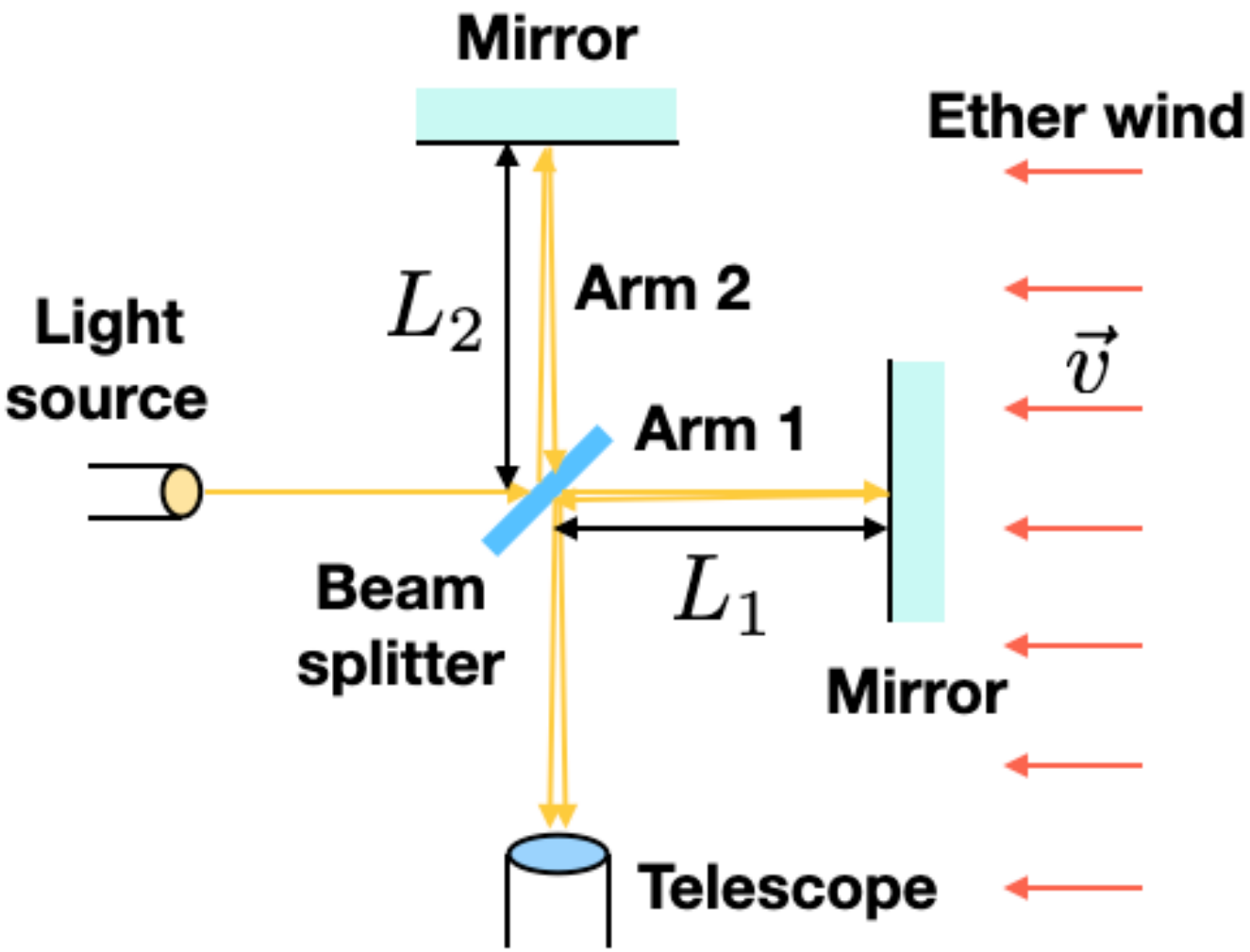
- Light is a transverse wave (Young and Fresnel, 1800s).
- Historically light is a mechanical wave which requireds a medium called *ether*.
- Wave equation of $\vec{E}(x, y, z)$ (or $\vec{B}(x, y, z)$) field

$$\nabla^2 \vec{E} = \frac{\partial^2 \vec{E}}{\partial x^2} + \frac{\partial^2 \vec{E}}{\partial y^2} + \frac{\partial^2 \vec{E}}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2}, \quad (1)$$

where c is the speed of light with respect to ether.



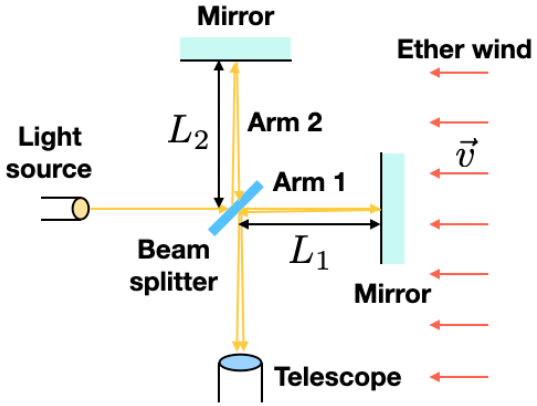
Michelson-Morley Experiment



Time for the round trip

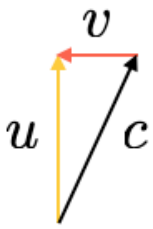
- Arm 1:

$$\begin{aligned}\Delta t_1 &= \Delta t_{1,\text{forward}} + \Delta t_{1,\text{backward}} \\ &= \frac{L_1}{u_{1,\text{forward}}} + \frac{L_1}{u_{1,\text{backward}}} \\ &= \frac{L_1}{c+v} + \frac{L_1}{c-v} = \frac{2L1/c}{(1-v^2/c^2)}. \end{aligned} \tag{2}$$



- Arm 2:
  We have $u_{2,\text{forward}} = u_{2,\text{forward}} = u = \sqrt{c^2 - v^2}$,

$$\begin{aligned}\Delta t_2 &= \Delta t_{2,\text{forward}} + \Delta t_{2,\text{backward}} \\ &= \frac{L_2}{u} + \frac{L_2}{u} = \frac{2L_2}{\sqrt{c^2-v^2}} = \frac{2L_2/c}{(1-v^2/c^2)^{1/2}}. \end{aligned} \tag{3}$$

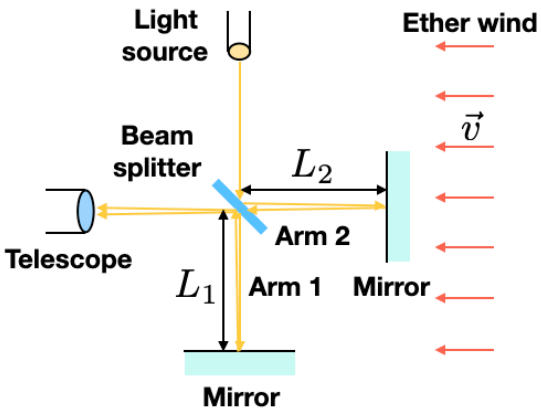


By using $(1+x)^n \approx 1 + nx$ for small x, the time difference (for $L_1 = L_2 = L_0$) is

$$\Delta T = \Delta t_1 - \Delta t_2 = \frac{L_0}{c}\left(\frac{v}{c}\right)^2. \tag{4}$$

- Rotate by $\pi/2$, $\Delta t_1' = \Delta t_2$ and $\Delta t_2' = \Delta t_1$,

$$\Delta T' = \Delta t_1' - \Delta t_2' = -\left(\frac{L_0}{c}\right)\left(\frac{v}{c}\right)^2. \tag{5}$$



- The rotation cause the fringe pattern (interference of light) to shift by

$$\Delta T - \Delta T' = \Delta t_1' - \Delta t_2' = \left(\frac{2L_0}{c}\right)\left(\frac{v}{c}\right)^2. \tag{6}$$

- If $v = 30$ km/s, then they expect to have 0.4 fringe. However, the result is nothing.
- They concluded that the speed of the earth relative to the ether had to be less than 5 km/s.
- Upper limit for v: 5 cm/s.

$$\Longrightarrow \quad \boxed{v = 0 \text{ for ether wind.}} \tag{7}$$

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1 Physical Observation

2 Special Relativity

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Special Relativity

- Galileo's relativity
  - Galileo transformation of coordinates

$$x' = x - ut \quad \Longrightarrow \quad v' = v - u. \tag{8}$$

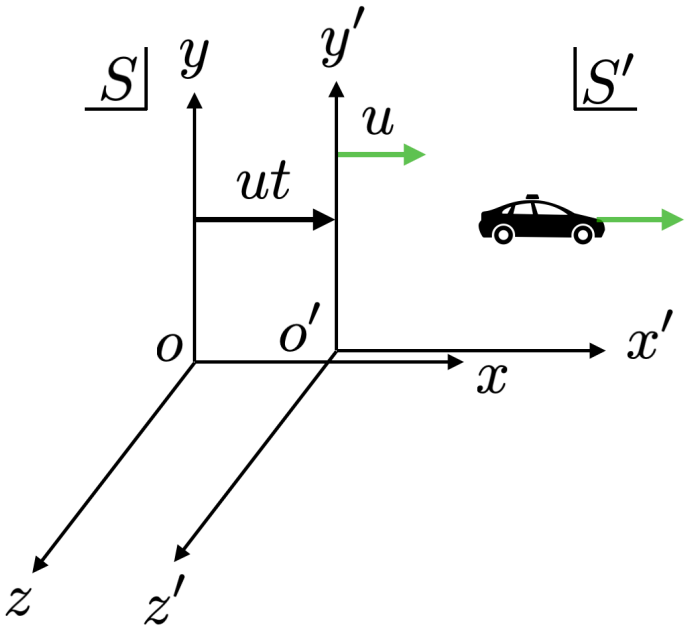
  - If $v = u$, then $v' = 0$ in S', we call S' the *rest frame* of the car.
- Relativity
  - Special theory (Lorentz, Poincaré, Einstein, etc.).

$$x' = \gamma(x - ut), \tag{9a}$$

$$t' = \gamma\Big(t - (u/c^2)x\Big). \tag{9b}$$

  - General theory (Einstein, Grossmann, Hilbert) includes the *gravitational effect*.

$$\underbrace{G_{\mu\nu}}_{\text{Geometry}} = 8\pi G \underbrace{T_{\mu\nu}}_{\text{Matter}} \tag{10}$$



Postulates of Special Theory of Relativity

- The Principle of Relativity (Galileo): All physical laws have the same form in all inertial frames.
- The Principle of the Constancy of the Speed of Light (Einstein): The speed of light in free space is the same in all inertial frames. It does not depend on the motion of the source or the observer.

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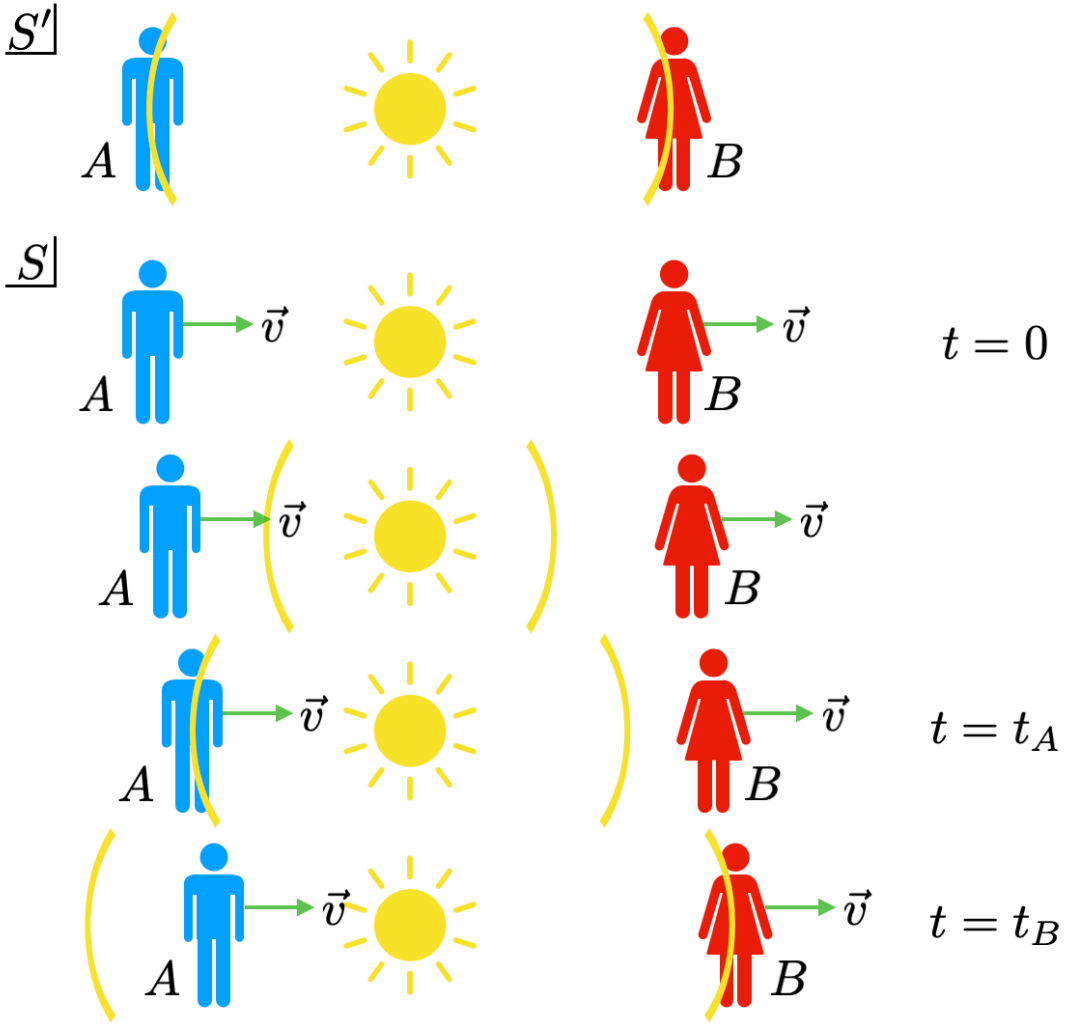
The Relativity of Simultaneity

- Simultaneity is relative.
  - A and B see the light simultaneously.
  - A and B do not see the light at the same time.
    $\Longrightarrow$ Light does not flash at the same time with respect to A and B.



Time Dilation

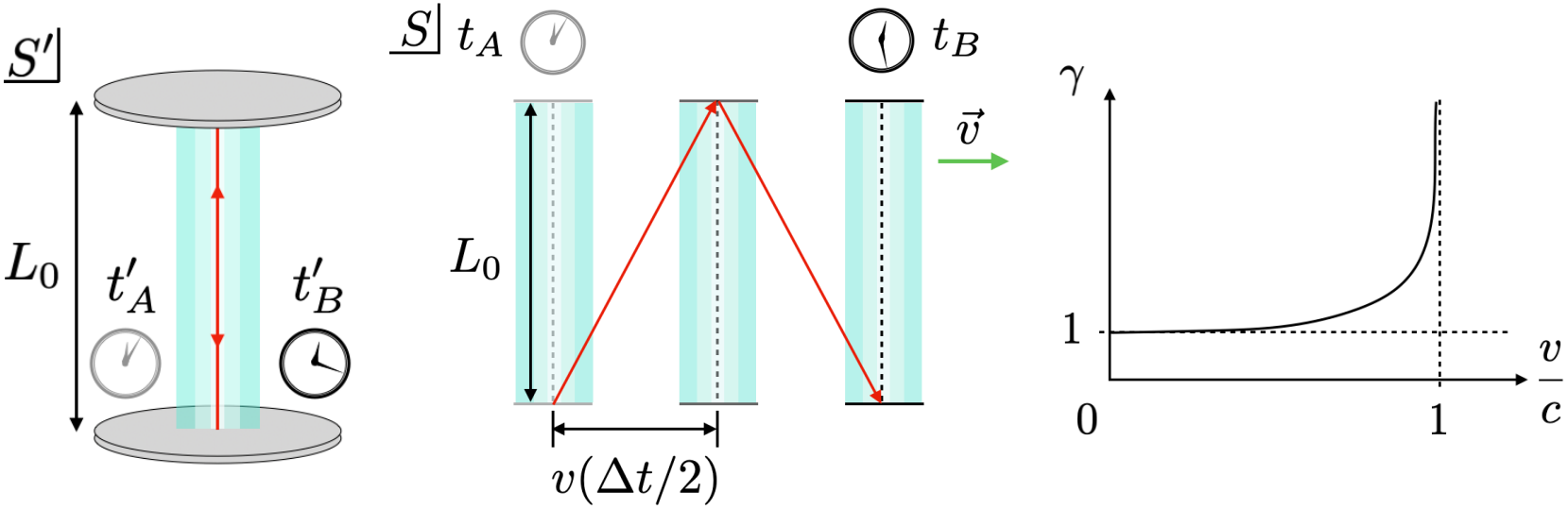
- In moving frame S' on the train, the round trip time of light is measured by observer at rest *at the same point*, which is called the *proper time*

$$T_0 = \Delta t' = \frac{2L_0}{c}. \tag{11}$$

- In the rest frame S, the train is moving with a velocity $\vec{v} = v\hat{i}$, the round trip time is

$$\left(\frac{c\Delta t}{2}\right)^2 = L_0^2 + \left(\frac{v\Delta t}{2}\right)^2 \quad \Longrightarrow \quad T = \Delta t = \frac{2L_0}{c}\left(\frac{1}{\sqrt{1 - v^2/c^2}}\right) \equiv \gamma T_0.$$

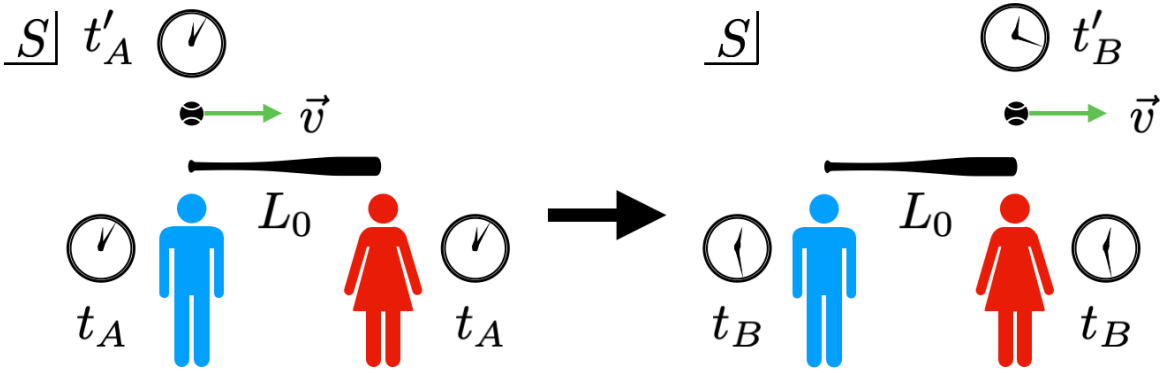
- Define the *Lorentz (gamma) factor* $\boxed{\gamma = \frac{1}{\sqrt{1-\beta^2}} \geq 1}$ with $\boxed{\beta = v/c} \leq 1$.
- $T \geq T_0 \Longrightarrow$ *Time dilation* (for $v \neq 0$).



Length Contraction

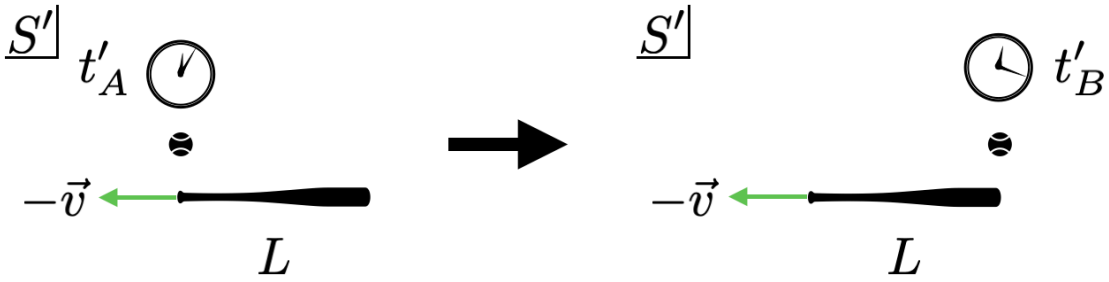
- In the rest frame S on the train, the length can be measured by people *at rest* through the time interval, which is called the *proper length*

$$L_0 = \Delta x = v\,\Delta t. \tag{12}$$



- In the moving frame S', the length is determined by the time interval of a *moving* clock

$$L = \Delta x' = v\,\Delta t' = v\left(\frac{\Delta t}{\gamma}\right) = \frac{L_0}{\gamma} \tag{13}$$



- $L \leq L_0 \implies$ *Length contraction* (for $v \neq 0$).

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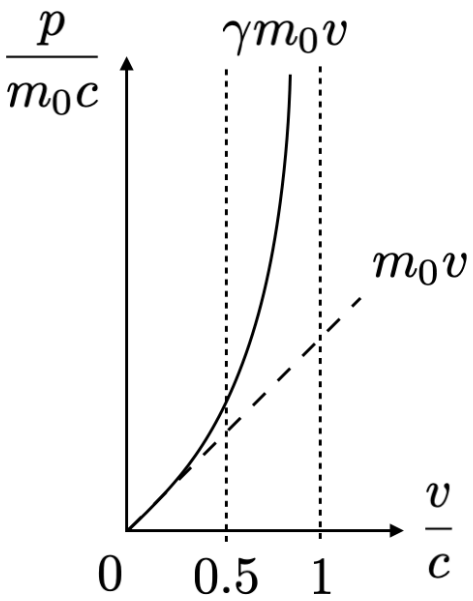
Relativistic Dynamics

- m_0 called the *rest mass* of a particle is measured in its *rest frame*.
- The relativistic mass is $m = \gamma m_0 = \dfrac{m_0}{\sqrt{1 - v^2/c^2}}$.
- The relativistic linear momentum is $mv = \gamma m_0 v$.
  - The speed of a *massive* particle cannot reach the light speed c.
  - Only *massless* particle can move with speed $v = c$, e.g., photon.
- The mass-energy equivalence $E = mc^2$ (the *total energy*).
- The *rest energy* is $m_0 c^2$
- Kinetic energy $K = E - m_0 c^2 = m_0 c^2(\gamma - 1)$
  - For $v/c \ll 1$ we have $\gamma \approx 1 + \dfrac{1}{2}\dfrac{v^2}{c^2}$, then $K \approx \dfrac{1}{2} m_0 v^2$.
  - If $K \ll E$, then $E = \gamma m_0 c^2 \approx E_0 = m_0 c^2$ and $\gamma \approx 1$ implying $m = m_0$.



Note

If $x \ll 1$, then we have the *binomial approximation*

$$(1 - x)^n \approx 1 - nx. \tag{14}$$

Mass-Energy relationship

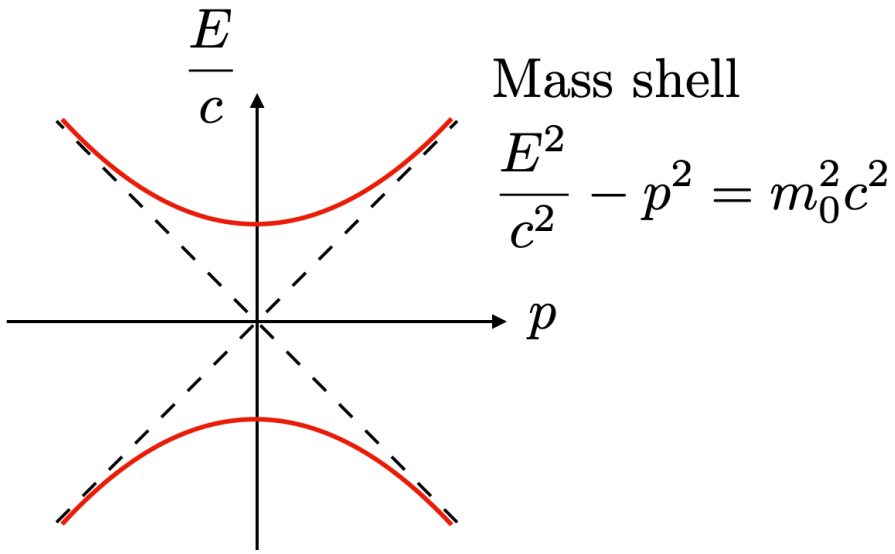
- From the relativistic definition of the energy, we have

$$E^2 - E_0^2 = (\gamma m_0 c^2)^2 - (m_0 c^2)^2 = m_0^2 c^4 \underbrace{(\gamma^2 - 1)}_{\frac{1}{1-v^2/c^2} - 1 = \frac{v^2/c^2}{1-v^2/c^2} = \gamma^2 v^2/c^2} = (\gamma^2 m_0^2 v^2) c^2 = p^2 c^2. \quad (15)$$

Therefore we obtain the mass-energy relationship which is invariant for any particle in special relativit

$$\boxed{E^2 = p^2 c^2 + m_0^2 c^4.} \quad (16)$$

- The mass shell: $E^2/c^2 - p^2 = m_0^2 c^2$.
- If $E \gg m_0 c^2$, then $E \approx pc$ for a particle whose speed is close to c.
- For phtons ($m_0 = 0$), we have $E = pc$.



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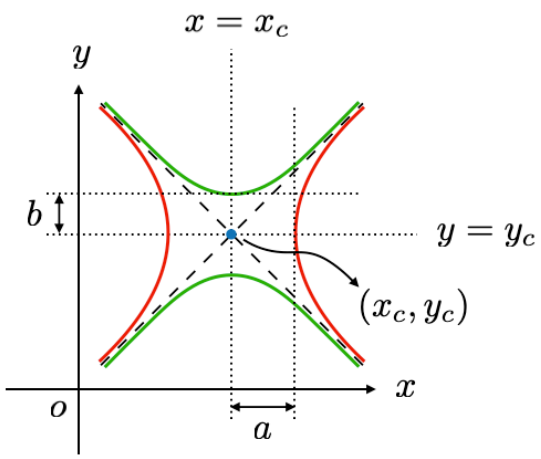
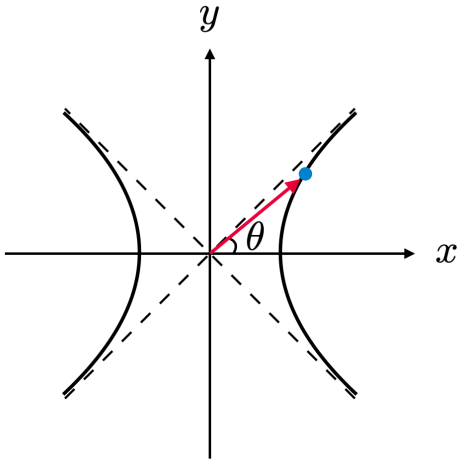
A hyperbolic space

- A hyperbola in two-dimensional plane

$$\frac{(x-x_c)^2}{a^2} - \frac{(y-y_c)^2}{b^2} = 1 \quad \text{or} \quad -\frac{(x-x_c)^2}{a^2} + \frac{(y-y_c)^2}{b^2} = 1 \tag{17}$$





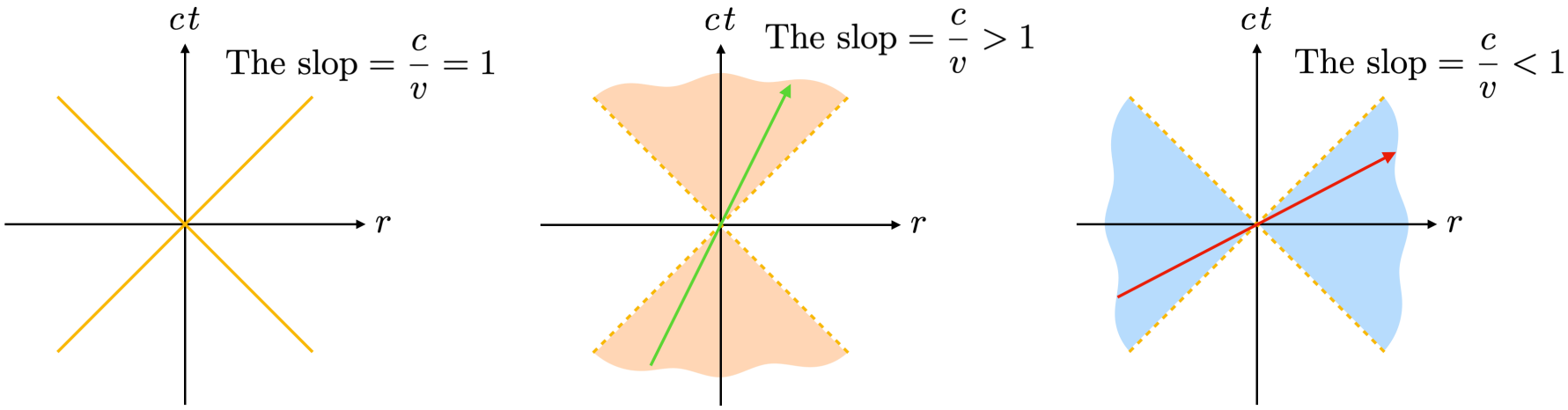
- Take $y = ct$, $x_c = 0 = y_c$ and $a = R = b$, the equation of a hyperbola becomes

$$\frac{(\Delta x)^2}{R^2} - \frac{(\Delta(ct))^2}{R^2} = 1 \quad \text{or} \quad x^2 - (ct)^2 = R^2, \tag{18}$$

which leads to a definition of the so-called *hyperbolic angle* θ. Hence the coordinates in terms of the hyperbolic angle θ are

$$\begin{cases} ct = R\sinh\theta, \\ x = R\cosh\theta. \end{cases} \tag{19}$$

Spacetime Diagram



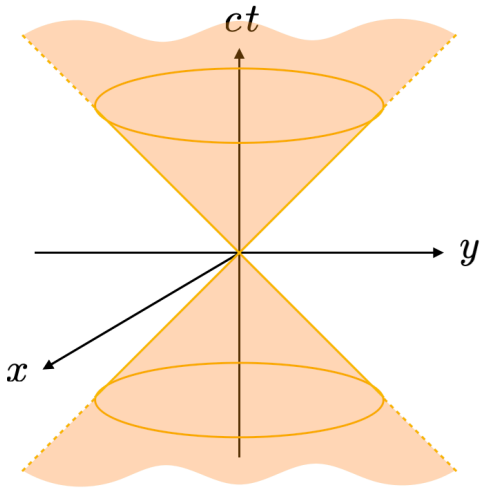
- Allowed region ($\frac{c}{v} \geq 1$)

Timelike: $(c\,dt)^2 > dr^2 \implies ds^2 \overset{\text{def}}{=} c^2\,dt^2 - dr^2 > 0,$ (20)

Lightlike: $(c\,dt)^2 = dr^2 \implies ds^2 \overset{\text{def}}{=} c^2\,dt^2 - dr^2 = 0.$ (21)

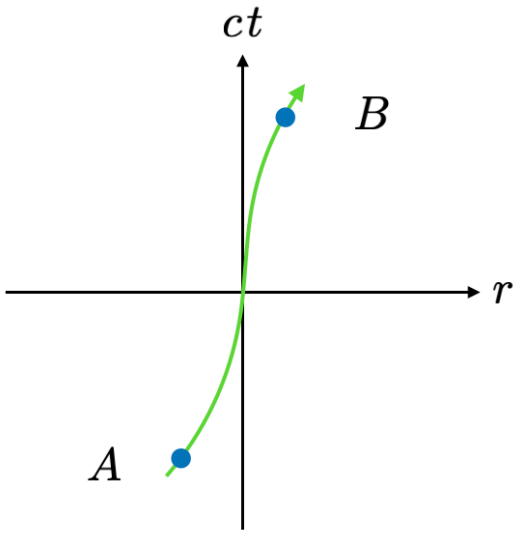
- Forbidden region ($\frac{c}{v} < 1$)

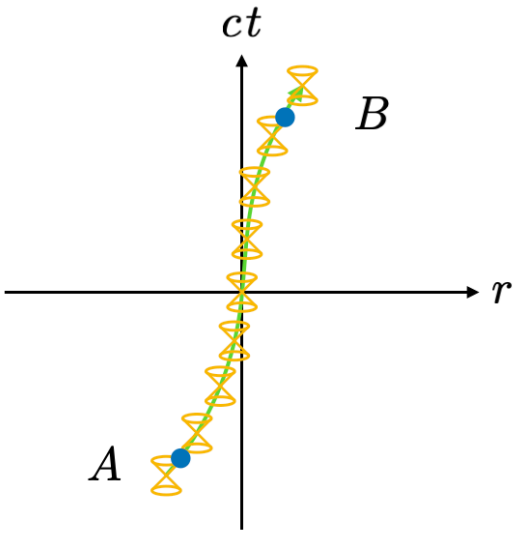
Spacelike: $(c\,dt)^2 < dr^2 \implies ds^2 \overset{\text{def}}{=} c^2\,dt^2 - dr^2 < 0.$ (22)

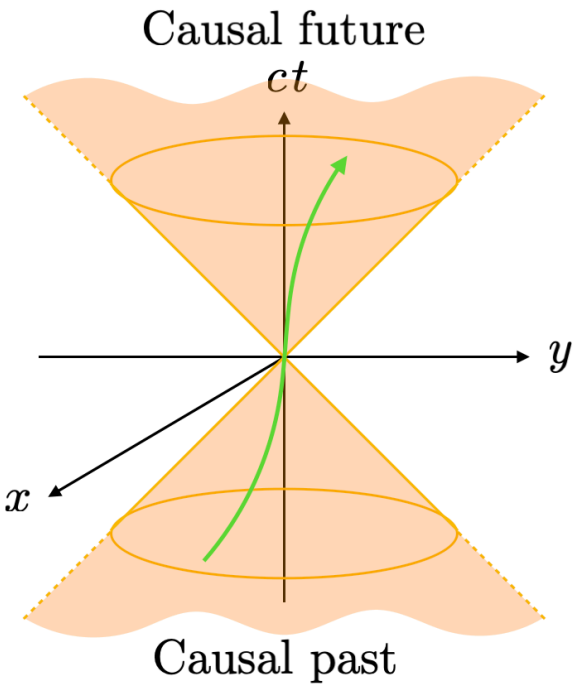


Causality and World Line

- Normal matter: $v \leq c \Longrightarrow$ the world line is *inside/along* the lightcone.
- A point determined by a time and a place in the spacetime diagram is calle an *event*.
- We say that events A and B are *causally related*.







- Only *tachyon* can travel in the *spacelike* region.

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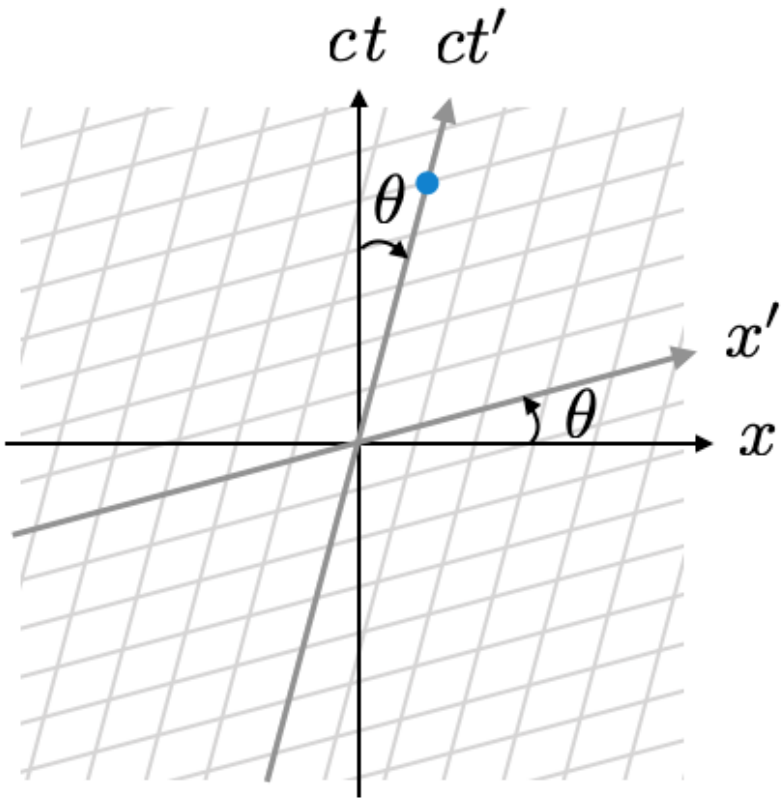
Lorentz Transformation

- Galileo's transformation is not valid when $v \to c$.
- The hyperbolic rotation

$$\begin{cases} ct' &= (\cosh\theta)(ct) - (\sinh\theta)x \\ &= \cosh\theta\big(ct - \tanh\theta x\big), \\ x' &= (-\sinh\theta)(ct) + (\cosh\theta)x \\ &= \cosh\theta\big(x - \tanh\theta(ct)\big), \\ y' &= y, \\ z' &= z. \end{cases} \tag{23}$$



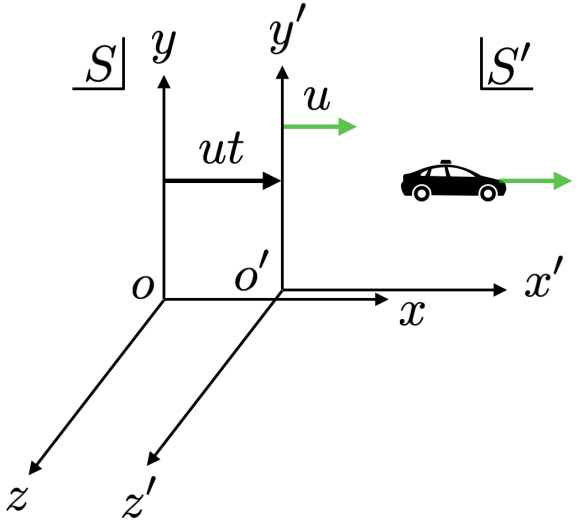
- Boost to the rest frame of the event $\implies x' = 0$, we have $\boxed{\beta = \frac{v}{c} = \tanh\theta.}$
- Use identity $\cosh\theta^2 - \sinh^2\theta = 1 \implies \boxed{\cosh\theta = \frac{1}{\sqrt{1-\tanh^2\theta}} = \gamma.}$

- The *Lorentz coordinate transformation* ($S \to S'$)

$$\begin{cases} ct' = \gamma(ct - \beta x), \\ x' = \gamma(x - \beta ct), \\ y' = y, \\ z' = z. \end{cases} \Longrightarrow \begin{cases} c\,dt' = \gamma\big(c\,dt - (u/c)\,dx\big) = \gamma c\,dt(1 - uv_x/c^2), \\ dx' = \gamma\big(dx - (u/c)\,c\,dt\big) = \gamma\,dt(v_x - u), \\ dy' = dy = v_y\,dt, \\ dz' = dz = v_z\,dt. \end{cases} \tag{24}$$

- The *relativistic velocity-addition formula* ($S \to S'$)

$$\begin{cases} v'_x = \dfrac{dx'}{dt'} = \dfrac{v_x - u}{1 - uv_x/c^2}, \\ v'_y = \dfrac{dy'}{dt'} = \dfrac{v_y}{\gamma(1 - uv_x/c^2)}, \\ v'_z = \dfrac{dz'}{dt'} = \dfrac{v_z}{\gamma(1 - uv_x/c^2)}. \end{cases} \tag{25}$$

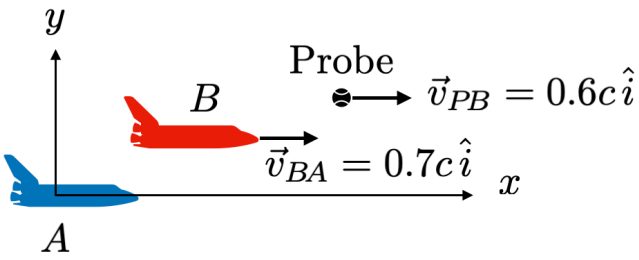


- For $S' \to S$, we need $u \to -u$ ($\beta \to -\beta$).

Example

- By Galilean formula $v_{PA} = v_{PB} + v_{BA} = 1.3c > c$.
- Using the relativistic formula

$$v_{PA} = \frac{v_{PB} + v_{BA}}{1 + v_{PB}v_{BA}/c^2} = 0.92c. \tag{26}$$



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Outline

7 Backup Slides

General Relativity

- A line element of a $(3+1)$-D spacetime $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 \equiv -\overbrace{d\tau^2}^{\text{proper time}}$.
- Generalized coordinate transformation $\{t, x, y, z\} \to \{\alpha, \beta, \gamma, \delta\}$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} A^\alpha{}_t & A^\alpha{}_x & A^\alpha{}_y & A^\alpha{}_z \\ A^\beta{}_t & A^\beta{}_x & A^\beta{}_y & A^\beta{}_z \\ A^\gamma{}_t & A^\gamma{}_x & A^\gamma{}_y & A^\gamma{}_z \\ A^\delta{}_t & A^\delta{}_x & A^\delta{}_y & A^\delta{}_z \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{\partial x^\alpha}{\partial x^t} & \frac{\partial x^\alpha}{\partial x^x} & \frac{\partial x^\alpha}{\partial x^y} & \frac{\partial x^\alpha}{\partial x^z} \\ \frac{\partial x^\beta}{\partial x^t} & \frac{\partial x^\beta}{\partial x^x} & \frac{\partial x^\beta}{\partial x^y} & \frac{\partial x^\beta}{\partial x^z} \\ \frac{\partial x^\gamma}{\partial x^t} & \frac{\partial x^\gamma}{\partial x^x} & \frac{\partial x^\gamma}{\partial x^y} & \frac{\partial x^\gamma}{\partial x^z} \\ \frac{\partial x^\delta}{\partial x^t} & \frac{\partial x^\delta}{\partial x^x} & \frac{\partial x^\delta}{\partial x^y} & \frac{\partial x^\delta}{\partial x^z} \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}$$

and the line element becomes $ds^2 = \sum_{\alpha,\beta} g_{\alpha\beta}\, dx^\alpha\, dx^\beta$.

- *Geodesic equation*: An equation of motion of a *free* (no external non-gravitaional forces) test particle

$$\frac{d^2 x^i}{d\tau^2} = 0 = f^i \quad \Longrightarrow \quad \boxed{\frac{d^2 x^\alpha}{d\tau^2} = \underbrace{\Gamma^\alpha{}_{\beta\gamma}}_{\text{Christoffel symbol}} \frac{dx^\beta}{d\tau}\frac{dx^\gamma}{d\tau} = \underbrace{f^\alpha}_{\text{Gravity force}} \quad \text{Einstein } \textit{Equivalence Principle}}.$$

where $\Gamma^\alpha{}_{\beta\gamma} = \dfrac{\partial^2 x^i}{\partial x^\beta \partial x^\gamma}\dfrac{\partial x^\alpha}{\partial x^i} = \left(\dfrac{\partial}{\partial x^\gamma} A^i{}_\beta\right) A^\alpha{}_i$.

- Reimaanian curvature $R^\alpha{}_{\beta\gamma\delta} = \frac{\partial}{\partial x^\gamma}\Gamma^\alpha{}_{\beta\delta} - \frac{\partial}{\partial x^\delta}\Gamma^\alpha{}_{\beta\gamma} + \Gamma^\alpha{}_{\lambda\gamma}\Gamma^\lambda{}_{\beta\delta} + \Gamma^\alpha{}_{\lambda\delta}\Gamma^\lambda{}_{\beta\gamma}$.
- *Einstein equation* (Einstein, Grossmann, Hilbert)

$$\boxed{\underbrace{G_{\alpha\beta}}_{\text{Geometry}} = 8\pi G \underbrace{T_{\alpha\beta}}_{\text{Energy-momentum tensor of matters}} \text{ where } G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R \text{ and } T_{\alpha\beta} = -\frac{1}{2}\frac{1}{\sqrt{\det g_{\alpha\beta}}}\frac{\delta S_{\text{matter}}}{\delta g^{\alpha\beta}}}.$$